Environmental Engineering (EE)

QUIZ 1 FOR FA1

1. In an open well recuperation test, what does the specific capacity factor Ks = (A/T) × ln(H1/H2) physically represent?

A. Maximum drawdown permissible before cavity collapse
B. Inflow volume per unit time entering the well under 1 m head
C. Total discharge of the well under a unit depression head
D. Rate of water inflow per unit surface area of well lining
Ans. B. Inflow volume per unit time entering the well under 1 m head

2. A pumping well draws water from an aquifer of constant saturated thickness 20 m, and the aquifer is bounded by confining layers. Which approach is more appropriate?

A. Thiem confined-aquifer equation
B. Freeman equation
C. Dupuit unconfined equation
D. Arithmetic population method
Ans. A. Thiem confined-aquifer equation

3. Which variable is particularly important in the confined-aquifer expression?

A. Aquifer thickness
B. Population growth rate
C. Per-capita demand
D. Fire-flow coefficient
Ans. A. Aquifer thickness

4. A geological stratum has high porosity due to microscopic voids but extremely low to no hydraulic conductivity, preventing appreciable flow under normal hydraulic gradients. This stratum is classified as:

A. Aquifer
B. Aquiclude
C. Aquitard
D. Aquifuge
Ans. B. Aquiclude (the quiz's selected Aquitard was marked incorrect)

5. For a design population of 100,000 with an average water demand of 250 LPCD, determine the coincident draft given that the fire demand is 0.65 m^3/s.

A. 0.289 m^3/s
B. 1.171 m^3/s
C. 0.781 m^3/s
D. 0.521 m^3/s
Ans. B. 1.171 m^3/s

6. If the city adopts Freeman's formula to determine its fire demand instead of the design fire demand of 0.65 m^3/s, by how much does the resulting coincident draft decrease?

A. 0.390 m^3/s
B. It does not decrease; it remains governed by MHD
C. 0.082 m^3/s
D. 0.120 m^3/s
Ans. C. 0.082 m^3/s

7. A recuperation test on a 1.4 m diameter well shows that the water level rises from an initial depression head of 4.2 m to 1.8 m in 80 minutes. What is the specific capacity (Ks) of the well in m3/hr per metre head?

A. 0.0098 m3/hr
B. 1.178 m3/hr
C. 2.355 m3/hr
D. 0.978 m3/hr
Ans. D. 0.978 m3/hr per metre head

8. For the exact same well, how many minutes will it take to recuperate from a depression head of 5.3 m down to 1.2 m?

A. 140.25 minutes
B. 310.0 minutes
C. 180.2 minutes
D. 253.6 minutes
Ans. A. 140.25 minutes

9. The standard allowance for water losses and unaccounted-for theft in an urban distribution network is generally taken as:

A. 35% to 40% of total demand
B. 2% to 5% of total demand
C. 10% to 15% of total demand
D. 20% to 30% of total demand
Ans. C. 10% to 15% of total demand

10. During excavation, a water-bearing layer is found that can store and transmit groundwater. It is bounded above and below by relatively impermeable formations. What type of aquifer does this most likely represent?

A. Perched aquifer
B. Confined aquifer
C. Aquifuge
D. Unconfined aquifer
Ans. B. Confined aquifer

11. Which observation most strongly distinguishes it from an unconfined aquifer?

A. It has groundwater flow
B. It is bounded by confining layers
C. It has saturated material
D. It contains water
Ans. B. It is bounded by confining layers

12. Census data of a city shows: 1991: 40,000; 2001: 48,000; 2011: 58,000. What is the projected population in 2031 using the Incremental Increase method?

A. 76,000
B. 85,000
C. 80,000
D. 82,000
Ans. D. 82,000

13. If an engineer uses the Geometric Increase method with the arithmetic mean of percentage decadal growth (20.415%), what is the forecast difference compared to the Incremental Increase projection?

A. Underestimates by 3,550
B. Overestimates by 2,098

C. Overestimates by 3,420
D. Underestimates by 1,200
Ans. B. Overestimates by approximately 2,099 (geometric projection ≈ 84,099 versus incremental projection of 82,000)

14. In a city with an annual average daily demand of q, what is the theoretical Maximum Hourly Demand (MHD) when considering peak hourly variations?

A. 1.5 × q
B. 2.7 × q
C. 1.8 × q
D. 3.2 × q
Ans. B. 2.7 × q

15. An unconfined pumping test (H = 20 m) yields drawdowns s1 = 2 m and s2 = 0.5 m at r1 = 0.5 m and r2 = 25 m, respectively, under Q = 3.5 m^3/min. What is K in m/min?

A. 0.01815 m/min
B. 0.00940 m/min
C. 0.03210 m/min
D. 0.05430 m/min
Ans. A. 0.01815 m/min

16. What is the expected steady water table height (h3) at an observation well placed at r3 = 50 m?

A. 20.00 m
B. 19.82 m
C. 18.90 m
D. 19.45 m
Ans. A. 20.00 m

17. What is the acceptable (desirable) limit of pH for drinking water as specified by IS 10500:2012?

A. 7.0 to 9.5
B. 5.5 to 7.0
C. 6.5 to 8.5
D. 6.0 to 9.0
Ans. C. 6.5 to 8.5

18. A town has a population of 1.80 lakh and an adopted water demand of 240 LPCD. Determine the maximum hourly demand in m3/s.

A. 1.200 m3/s
B. 0.900 m3/s
C. 0.750 m3/s
D. 1.080 m3/s
Ans. 1.35 m3/s using MHD = 2.7 × average daily demand; none of the listed options matches.

19. An area of 50 hectares receives a storm of 80 mm precipitation. Total infiltration loss is 35 mm and evaporation is 5 mm. What is the runoff coefficient C?

A. 62.5%
B. 40.0%
C. 37.5%
D. 50.0%
Ans. D. 50.0% (runoff depth = 80 − 35 − 5 = 40 mm; C = 40/80)

20. If urbanization cuts infiltration in half (to 17.5 mm) while evaporation remains unchanged, what is the new total runoff volume in m3?

A. 35,000 m3
B. 20,000 m3
C. 28,750 m3
D. 18,750 m3
Ans. C. 28,750 m3 (runoff depth = 80 − 17.5 − 5 = 57.5 mm; volume = 0.0575 × 500,000)

21. The color of drinking water on the platinum-cobalt scale is reported in which unit?

A. Hazen units
B. mg/L
C. TON
D. NTU
Ans. A. Hazen units

22. Total hardness in water is primarily caused by the dissolved multivalent cations of which two elements?

A. Copper and zinc
B. Calcium and magnesium
C. Iron and manganese
D. Sodium and potassium
Ans. B. Calcium and magnesium

23. Which apparatus is commonly used in the laboratory to measure the turbidity of a water sample by light scattering?

A. BOD incubator
B. Nephelometer
C. Tintometer
D. Osmoscope
Ans. B. Nephelometer

24. A well fully penetrates a confined aquifer of thickness 18 m with K = 45 m/day, R = 180 m, r = 0.18 m, and drawdown = 4 m. What is the discharge yield in m^3/day?

A. 3683.81 m^3/day
B. 5894.10 m^3/day
C. 2947.05 m^3/day
D. 1473.52 m^3/day
Ans. C. 2947.05 m^3/day

25. If the piezometric surface drops permanently below the top confining bed, transitioning the upper boundary into a water table, which equation replaces this formula?

A. Thiem's modified confined equation with variable transmissivity
B. Dupuit's unconfined equation using ($H^2 - h^2$)
C. Kuchling's hydraulic draft formula
D. The open well recuperation equation
Ans. B. Dupuit's unconfined equation using ($H^2 - h^2$)

26. According to IS 10500:2012, what is the maximum permissible limit for Total Dissolved Solids (TDS) in drinking water in the complete absence of an alternate source?

A. 1500 mg/L
B. 500 mg/L
C. 2000 mg/L
D. 1000 mg/L
Ans. C. 2000 mg/L